AB 100: CAPSTONE PROJECT-1

Enterprise Multi-Agent AI Business Platform

Project Overview

Design and implement an enterprise-grade Agentic AI platform using:

- Azure AI Foundry
- Azure OpenAI
- Copilot Studio
- Microsoft Graph API
- Power Automate
- SharePoint
- Dataverse
- Azure AI Search
- RBAC & Entra ID
- Enterprise AI Governance

The solution should simulate a real enterprise environment where multiple AI agents collaborate to automate business operations securely and intelligently.

Enterprise Business Scenario

A multinational enterprise wants to modernize operations using Agentic AI.

The organization currently faces:

- Slow manual workflows
- High operational costs
- Delayed approvals
- IT support overload
- Poor employee onboarding experience
- Disconnected enterprise systems
- Lack of AI governance

- No intelligent automation

The company wants to build a centralized AI business platform using Microsoft AI technologies.

Capstone Architecture

Core Components

AI Services

- Azure OpenAI
- Azure AI Foundry
- Azure AI Search

AI Agents

- HR AI Agent
- IT Support Agent
- Finance Approval Agent
- Procurement Agent
- Executive Assistant Agent

Enterprise Systems

- SharePoint
- Dataverse
- SQL Database
- Outlook
- Teams

Automation Layer

- Power Automate
- Event-driven workflows
- Approval workflows

Security Layer

- Azure Entra ID
- RBAC
- Managed Identities
- API Security

Monitoring Layer

- Azure Monitor
- AI Analytics
- Token Monitoring
- Governance Dashboard

Enterprise AI Agents Included

AI Agent	Purpose
HR AI Assistant	Employee onboarding & HR queries
IT Helpdesk Agent	Incident triage & troubleshooting
Finance AI Agent	Approval workflows & financial validation
Procurement Agent	Purchase approval automation
Executive AI Assistant	Meeting scheduling & email summarization
Knowledge Search Agent	Enterprise RAG search
Governance Agent	AI policy monitoring
Notification Agent	Teams & Outlook alerts

CASE STUDIES / PROBLEM STATEMENTS

Case 1 — Enterprise HR AI Assistant

Problem Statement

Employees frequently raise repetitive HR queries:

- Leave policy
- Payroll issues
- Insurance benefits
- Onboarding process

HR teams spend excessive time responding manually.

Solution Approach

- Build HR AI Assistant using Copilot Studio
- Ground AI using SharePoint HR documents
- Use RAG architecture
- Integrate Microsoft Graph
- Enable Teams chatbot access

Key Features

- HR FAQ automation
- Leave balance lookup
- Policy retrieval
- Employee onboarding guidance
- Adaptive cards

Expected Outcome

- Reduced HR workload
- Faster employee support
- Improved onboarding experience

Case 2 — AI-Powered IT Helpdesk Automation

Problem Statement

IT support teams receive thousands of repetitive tickets:

- Password reset
- VPN issues
- Teams problems
- Outlook crashes

Ticket routing is manual and slow.

Solution Approach

- Create IT Support Agent
- Use AI classification
- Implement intelligent ticket routing
- Integrate ServiceNow/Dataverse
- Use Power Automate escalation workflows

Key Features

- Ticket classification
- Severity detection
- AI troubleshooting
- Auto escalation
- Teams notifications

Expected Outcome

- Faster incident resolution
- Reduced support workload
- Improved SLA compliance

Case 3 — Finance Approval Automation

Problem Statement

Financial approvals are delayed due to:

- Manual verification

- Long email chains
- Policy validation delays

Solution Approach

- Build Finance Approval AI Agent
- Integrate Power Automate approvals
- Validate against company finance policies
- Use AI reasoning for risk analysis

Key Features

- Expense validation
- AI-based approval recommendations
- Conditional approvals
- Approval audit tracking

Expected Outcome

- Faster approvals
- Reduced manual effort
- Improved financial governance

Case 4 — AI-Based Executive Assistant

Problem Statement

Executives spend excessive time managing:

- Emails
- Meetings
- Scheduling
- Summaries

Solution Approach

- Integrate Microsoft Graph APIs
- Build AI Executive Assistant
- Use Outlook & Teams integration
- Automate meeting summaries

Key Features

- Calendar scheduling
- Email summarization
- Action item extraction
- Meeting assistant

Expected Outcome

- Increased executive productivity
- Better meeting management
- Reduced administrative effort

Case 5 — Enterprise Knowledge Search using RAG

Problem Statement

Employees cannot easily locate:

- SOPs
- Policies
- Technical documents
- Internal procedures

Solution Approach

- Build RAG-based enterprise search
- Use Azure AI Search
- Implement embeddings
- Ground AI responses on enterprise documents

Key Features

- Semantic search
- Context-aware AI responses
- SharePoint grounding
- Enterprise knowledge retrieval

Expected Outcome

- Faster knowledge discovery
- Reduced repetitive support requests
- Improved employee productivity

Case 6 — AI Procurement Workflow

Problem Statement

Procurement approvals involve:

- Multiple stakeholders
- Vendor verification
- Budget validation
- Delayed approvals

Solution Approach

- Create Procurement AI Agent
- Integrate approval workflows
- Use AI risk analysis
- Automate notifications

Key Features

- Vendor verification
- Budget analysis

- Approval orchestration
- Multi-agent workflows

Expected Outcome

- Faster procurement cycle
- Improved compliance
- Better approval visibility

Case 7 — Intelligent Ticket Routing

Problem Statement

Support tickets are routed incorrectly leading to:

- Delays
- Reassignments
- SLA violations

Solution Approach

- Use AI classification
- Detect business context
- Route tickets intelligently
- Integrate Teams notifications

Key Features

- AI categorization
- Priority detection
- Department-based routing
- Automated escalation

Expected Outcome

- Improved routing accuracy

- Faster ticket handling
- Reduced manual triage

Case 8 — AI Governance & Responsible AI

Problem Statement

Enterprise AI deployments lack:

- Governance
- Monitoring
- Auditability
- Responsible AI controls

Solution Approach

- Implement AI governance framework
- Configure RBAC
- Enable content filtering
- Monitor token usage
- Implement approval checkpoints

Key Features

- AI audit logging
- Prompt filtering
- Governance dashboards
- Role-based security

Expected Outcome

- Secure AI adoption
- Responsible AI compliance
- Enterprise governance visibility

Case 9 — AI-Driven Enterprise Notifications

Problem Statement

Critical business notifications are delayed due to manual processes.

Solution Approach

- Build AI Notification Agent
- Use Power Automate
- Integrate Teams & Outlook

Key Features

- Real-time alerts
- Approval notifications
- Incident escalation
- Enterprise messaging

Expected Outcome

- Faster communication
- Improved operational efficiency
- Better incident response

Case 10 — Enterprise Multi-Agent AI Orchestration

Problem Statement

Departments operate independently without intelligent workflow coordination.

Solution Approach

- Build parent-child multi-agent architecture
- Implement orchestration engine
- Enable distributed AI workflows

Key Features

- Multi-agent collaboration
- Task delegation
- Workflow memory
- Cross-department automation

Expected Outcome

- Enterprise-wide automation
- Intelligent workflow orchestration
- Reduced operational silos

Enterprise Implementation Steps

Phase 1 — Environment Setup

Tasks

- Create Azure AI Foundry Workspace
- Deploy Azure OpenAI models
- Configure Azure AI Search
- Setup Entra ID security
- Configure RBAC

Phase 2 — AI Agent Development

Tasks

- Create Copilot Studio agents
- Configure prompts
- Setup conversation flows
- Configure adaptive cards

Phase 3 — Knowledge Grounding

Tasks

- Upload enterprise documents
- Configure RAG architecture
- Create embeddings
- Enable semantic search

Phase 4 — API & Integration Layer

Tasks

- Integrate Microsoft Graph APIs
- Configure Power Automate
- Connect SQL/Dataverse
- Setup enterprise connectors

Phase 5 — Automation Workflows

Tasks

- Create approval workflows
- Configure event-driven automation
- Enable notifications
- Implement escalations

Phase 6 — Security & Governance

Tasks

- Configure Entra ID
- Implement RBAC
- Enable monitoring
- Configure Responsible AI policies

Phase 7 — Monitoring & Optimization

Tasks

- Monitor token consumption

- Analyze AI usage
- Optimize prompts
- Tune performance

Enterprise Deliverables

Deliverable	Description
AI Architecture Diagram	End-to-end enterprise architecture
AI Agents	Multiple enterprise AI agents
RAG Implementation	Knowledge grounding system
Automation Workflows	Power Automate flows
Governance Model	Security & compliance implementation
Monitoring Dashboard	AI analytics & token monitoring
Final Presentation	Enterprise solution walkthrough

Final Enterprise Architecture

Users
↓
Copilot Studio
↓
API Management
↓
Azure OpenAI
↓
Azure AI Search
↓
Dataverse / SQL / SharePoint
↓
Power Automate
↓
Enterprise Systems

Recommended Technologies

Technology	Purpose
Azure OpenAI	LLM inference
Azure AI Foundry	AI orchestration
Copilot Studio	AI agents
Azure AI Search	RAG retrieval
Power Automate	Workflow automation
Microsoft Graph	Teams/Outlook integration
Dataverse	Business data
SharePoint	Knowledge grounding
Azure Monitor	Monitoring
Entra ID	Authentication

Final Conclusion

This capstone project provides:

- Real enterprise AI implementation experience
- Multi-agent architecture design
- Enterprise-grade automation skills
- RAG implementation expertise
- AI governance and security exposure
- Hands-on Microsoft AI ecosystem experience
- Production-ready AI solution design knowledge